

enhance the energy density of the battery. However, their application has been limited by factors such as cost and brittleness, which are being addressed through innovations like polymer encapsulation.

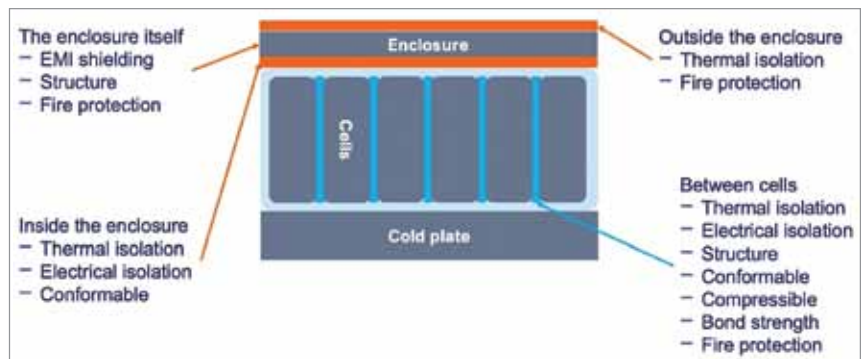
The industry relies on a variety of material suppliers to meet these evolving needs. For example, Aspen and IBIH are key players in the aerogel market, while Dow and HB Fuller provide encapsulating foams. Other suppliers, such as Asheville Mica and Morgan Advanced Materials, offer mica and ceramic blankets, respectively. This diverse supplier landscape is instrumental in advancing the safety and performance of EV batteries.

Cell-to-pack designs

Cell-to-pack (CTP) designs in EVs are an innovative approach to battery configuration, where individual cells are directly assembled into a pack, eliminating the need for modules. This design simplifies the manufacturing process and reduces costs but also presents challenges in fire protection. Since modules are no longer present, fire protection must be applied at the cell or pack level.

In CTP designs, the materials used for fire protection are critical, as they generally need to be placed between the cells. As manufacturers aim to make battery packs less complex and lighter, the ideal fire protection materials should serve multiple functions. They should not only provide fire protection but also offer thermal and electrical insulation. This multifunctionality means that fewer individual materials are required, leading to a more streamlined and efficient design.

When it comes to thermal management and safety, CTP designs require a different approach. Safety materials are more commonly placed between cells and at the pack level, either above or below the



Low thermal conductivity and density are just two of the key factors for fire protection materials in EV batteries (Source: IDTechEx)

cells. This is a significant shift from traditional designs that often relied on module-level protection. For thermal management, keeping cells at their optimal operating temperature is still a priority. The construction may be simpler, as seen in the BYD Blade battery pack, which uses a single large cold plate across all cells. Tesla's 4680 battery pack, with fewer individual cells, can utilise fewer coolant channels. However, the overall thermal management strategy remains similar to previous designs.

With the advent of CTP designs, the focus is on developing fire protection materials that are multi-functional, providing not only safety but also contributing to the thermal management of the battery pack. This approach is crucial in ensuring the safety and efficiency of EVs as they continue to evolve.

How much is just right

Estimating the amount of fire protection material required in EV batteries is a complex process that involves analysing the specific battery design and meeting specific performance characteristics. This process helps determine the necessary quantity of materials like aerogels, considering the space available for fire protection.

One challenge in this estimation is the variability in material thick-

ness and dimensions across different manufacturers, leading to some tolerance in the calculations. For example, while a 1.5-millimetre thick aerogel may be adequate for some designs, others might require a slightly different thickness.

These estimations are crucial for enhancing EV battery safety, as they provide a basis for comparing the amount of material needed per kilowatt-hour of battery capacity. This comparison allows for assessing the trade-offs between material lightness and protection requirements. By evaluating the mass and cost, battery makers can identify the most cost-effective fire protection solution that meets performance standards.

The density of the fire protection material is another important consideration. Ideally, a material should be as light as possible while still offering adequate protection, as a lighter material contributes to higher energy density and greater range for the battery. However, the relationship between density and fire protection capability is not straightforward and depends on the specific properties and performance of the material.

Ultimately, the challenge for battery manufacturers lies in balancing cost, weight, and protection when selecting materials. Fire protection is typically considered in the later stages of battery design, after the cell format and overall structure have